

Department Overview

- **Department Name:** Department of Petroleum and Gas Engineering
- **Faculty:** Faculty of Engineering & Architecture
- **History:** Established in the Fall of 2004 as the pioneering department of the newly formed engineering faculty to address the industrial need for managing Balochistan's extensive natural oil and gas reserves (such as the Sui gas fields).
- **Accreditation:** The four-year undergraduate program is fully accredited by the **Pakistan Engineering Council (PEC)** and follows the Outcome-Based Assessment (OBA) guidelines.
- **Core Focus:** Equips students with scientific, mathematical, and computer-based skills for the exploration, drilling, appraisal, and optimal development of oil and gas assets.

Mission

The program cultivates industrious engineers for the national and international petroleum and gas industry and its related fields in education, business, and government by providing technical knowledge, skills, and values through academic inquiry, teaching, and technical services.

Program Educational Objectives (PEOs)

Graduates of the program are expected to achieve the following milestones within 3–5 years of graduation:

1. **PEO-1:** Address complex problems in fields like renewable and non-renewable energy and sustainable development by utilizing professional, teamwork, communication, and managerial skills.
2. **PEO-2:** Fulfill the requirements and necessary skills for higher academic studies and cutting-edge research.
3. **PEO-3:** Adopt a healthy and symbiotic environment with a strong commercial instinct, helping create new opportunities, especially in Balochistan.

Bachelor of Science in Petroleum & Gas Engineering (BS PGE)

Admission Requirements

- **Academic Background:** F.Sc (Pre-Engineering) from a recognized board or equivalent with at least **60% marks**, OR a Diploma of Associate Engineer (DAE) in a relevant field with at least **60% marks**.
- **Testing:** An entry test administered by NTS.

Degree Requirements

- **Total Credit Hours:** 136

- **Total Courses:** 44
- **Minimum Academic Standing:** CGPA $\geq$ 2.0
- **PLO Requirement:** Students must successfully pass all 12 standard Program Learning Outcomes (ranging from Engineering Knowledge and Problem Analysis to Ethics and Lifelong Learning).

Curriculum Plan (Program Schema by Semester)

Note: Credit hour structures denote (Theory + Lab = Total).

Semester	Course Code	Course Title	Prerequisite	Credit Hours
1st	PGE-102	Fundamentals of Petroleum Engineering	None	2+0=2
	GEO-201	Applied Geology	None	2+1=3
	HUM-101/112	Islamic Studies/Ethics	None	2+0=2
	MATHA 111	Applied Mathematics-I	None	3+0=3
	PHY-205 / HUM-163	Applied Physics / Functional English (English-I)	None / None	3+1=4 / 3+0=3
	ENGG 202L	Workshop Practice	None	0+2=2
2nd	HUM-268 / HUM-102	Communication Skills (Eng-II) / Pakistan	None / None	3+0=3 / 2+0=2

		Studies		
	MATHA-116	Applied Mathematics-II	Applied Mathematics-I	3+0=3
	CHE-102	Applied Chemistry	None	2+1=3
	ENGG-201 L	Engineering Drawing	None	0+2=2
	CS-113	Introduction to Computer Programming	None	2+1=3
3rd	HUM-276	Technical Report Writing & Pres. Skills (Eng-III)	None	2+0=2
	MATHA-211	Applied Mathematics-III	Applied Mathematics-II	3+0=3
	ENGG-206 / 211	Fluid Mechanics / Principles of Electrical Eng.	None / None	2+1=3 / 2+1=3
	PGE-306 / ENGG-215	Petroleum Geology & Geophysical Exploration / Thermo	None / None	3+0=3 / 2+1=3

4th	STAT-203 / GEOE-206	Applied Statistics / Mechanics of Materials	None / None	3+0=3 / 2+1=3
	PGE-303 / PGE-302	Drilling Engineering-I / Petrophysics	None / None	3+1=4 / 3+1=4
	HUM-206	Introduction to Sociology	None	3+0=3
5th	MATHA-212	Applied Numerical Methods	None	2+1=3
	PGE-305 / PGE-402	Properties of Reservoir Fluids / Drilling Engineering-II	None / Drilling Eng-I	3+1=4 / 3+1=4
	ENGG-322 / CHEE-313	Environment & Safety Management / Process Control	None / None	3+0=3 / 2+1=3
6th	PGE-311 / PGE-304	Reservoir Geomechanics / Well Logging	None / Petrophysics	2+0=2 / 2+1=3
	PGE-309 / PGE-308	Reservoir Engineering I / Petroleum Production	None / None	3+1=4 / 3+1=4

		Eng-I		
	CHEE-416	Petroleum Refinery Engineering	None	3+1=4
7th	PGE-408L	Final Year Project (Part I)	None	0+3=3
	MGMT-402	Project Planning and Management	None	2+0=2
	PGE-404 / PGE-405	Well Testing / Petroleum Production Engineering-II	None / Petroleum Prod-I	3+1=4 / 3+1=4
	PGE-312	Reservoir Simulation	None	3+1=4
8th	PGE-408L	Final Year Project (Part II)	None	0+3=3
	PGE-412 / ECON-105	Natural Gas Engineering / Engineering Economics	None / None	2+1=3 / 2+0=2
	PGE-407 / PGE-403	Principles of Enhanced Oil Recovery / Reservoir Eng-II	None / None	3+1=4 / 3+1=4